

$$\begin{array}{r} 108 \\ -99 \\ \hline 09 \end{array}$$

/ /

1)

a) $47 \bmod 13 = 8$

$$\begin{array}{r} 47 \overline{) 13} \\ 8 \overline{) 3} \\ \hline 39 \end{array}$$

$= 47 \equiv 8 \pmod{13}$

b) $(18+25) \bmod 7$
 $43 \bmod 7$

$$\begin{array}{r} 43 \overline{) 7} \\ 1 \overline{) 6} \end{array}$$

$= 43 \equiv 1 \pmod{7}$

c) $(9 \cdot 12) \bmod 11$
 $108 \bmod 11$

$$\begin{array}{r} 108 \overline{) 11} \\ 9 \overline{) 9} \end{array}$$

$= 108 \equiv 9 \pmod{11}$

	1	(1)	15. $x \equiv 1 \pmod{26}$
2) 6, 26	(2)	15, 26	2
3, 13	3	15, 13	3
1, 13	13	5, 13	5
1, 13	13	1, 13	13
		1	
6 mod 26			
não existe inverso mod		possui!	
Credeal			*Chegou a hora de testar.

$$26, 15 \mid 1$$

$$15 \cdot 1 = 15 \rightarrow 15 \div 26 = \text{resto } 15$$

$$15 \cdot 2 = 30 \rightarrow 30 \div 26 = \text{resto } 4$$

$$15 \cdot 3 = 45 \rightarrow 45 \div 26 = \text{resto } 19$$

$$15 \cdot 4 = 60 \rightarrow 60 \div 26 = \text{resto } 8$$

$$15 \cdot 5 = 75 \rightarrow 75 \div 26 = \text{resto } 23$$

$$15 \cdot 6 = 90 \rightarrow 90 \div 26 = \text{resto } 12$$

$$15 \cdot 7 = 105 \rightarrow 105 \div 26 = \text{resto } 1 //$$

logo: $15 \cdot x \equiv 1 \pmod{26}$

$$15 \cdot 7 \equiv 1 \pmod{26}$$

$$105 \equiv 1 \pmod{26} //$$

$$105 \mid 26$$

$$= 1 // 4$$

$$x = 7 //$$

3. 10101100

$\oplus$ 00111101

10010001

reversão: $\rightarrow$ 10010001

$\oplus$ 10101100

00111101